

Meeting Notes 17.09.

Optimization Pipeline

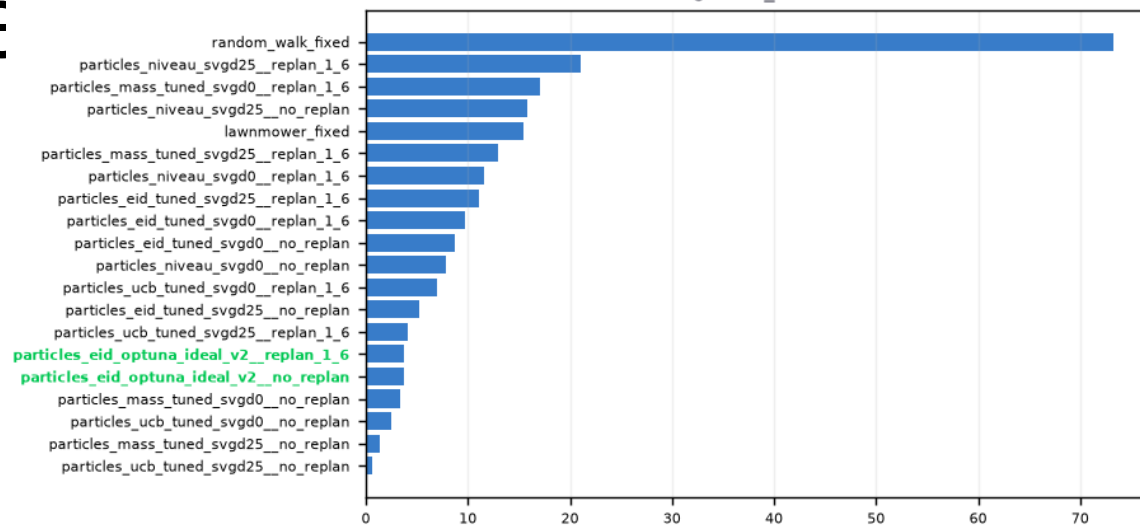
- Optimize $J = q + \lambda_{\text{len}} \cdot n + \lambda_{\text{time}} \cdot t$
- Searchspace:
 - phi_model: ucb, eid, mass, niveau
 - Param: kappa,w,tau
 - svgd_iters
 - debt_weight
- Used TPE (Tree-structured Parzen Estimator) with Optuna library
 - Bayes search, devide good and bad samples, create a density around them, calculate expected improvement and sample again
 - Reduces search space in comparison to combinatorial optimization by

E
p
re

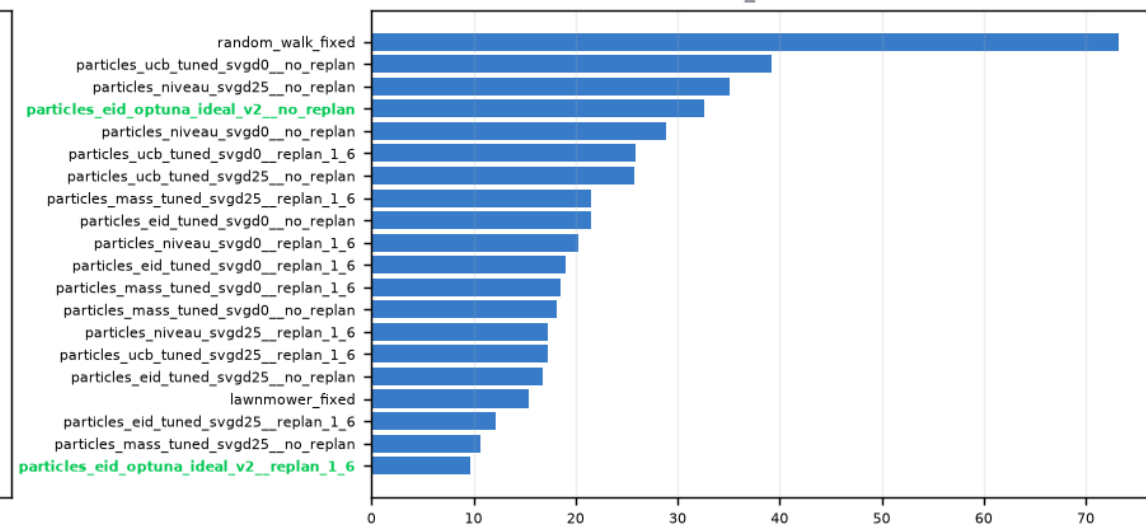
$$J = E_{\text{total}} + \lambda_{\text{len}} \cdot L(\gamma), \quad \lambda_{\text{len}} = 0.02$$

smaller = better

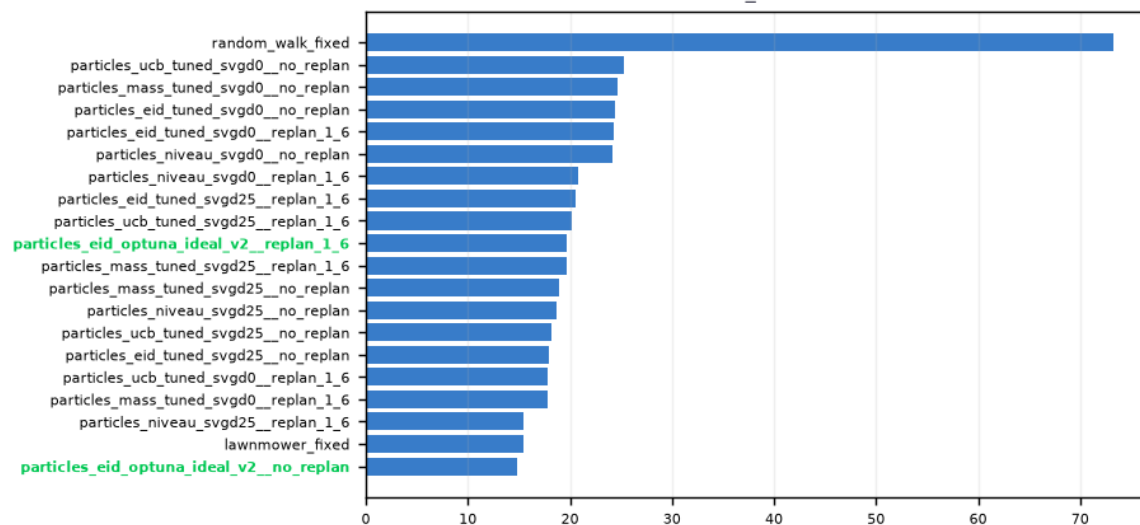
ground_truth



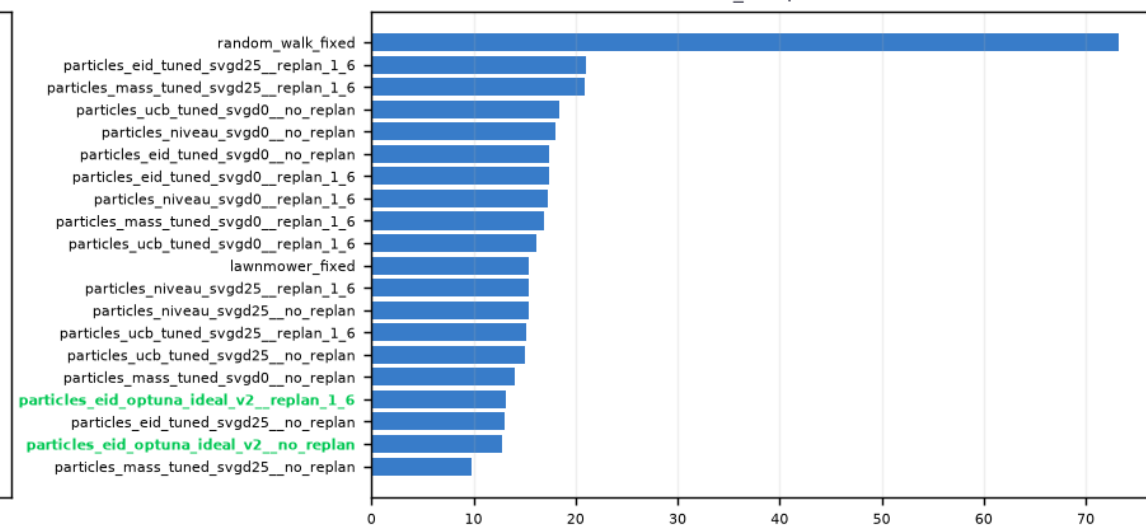
half_known



none_known



ten_samples

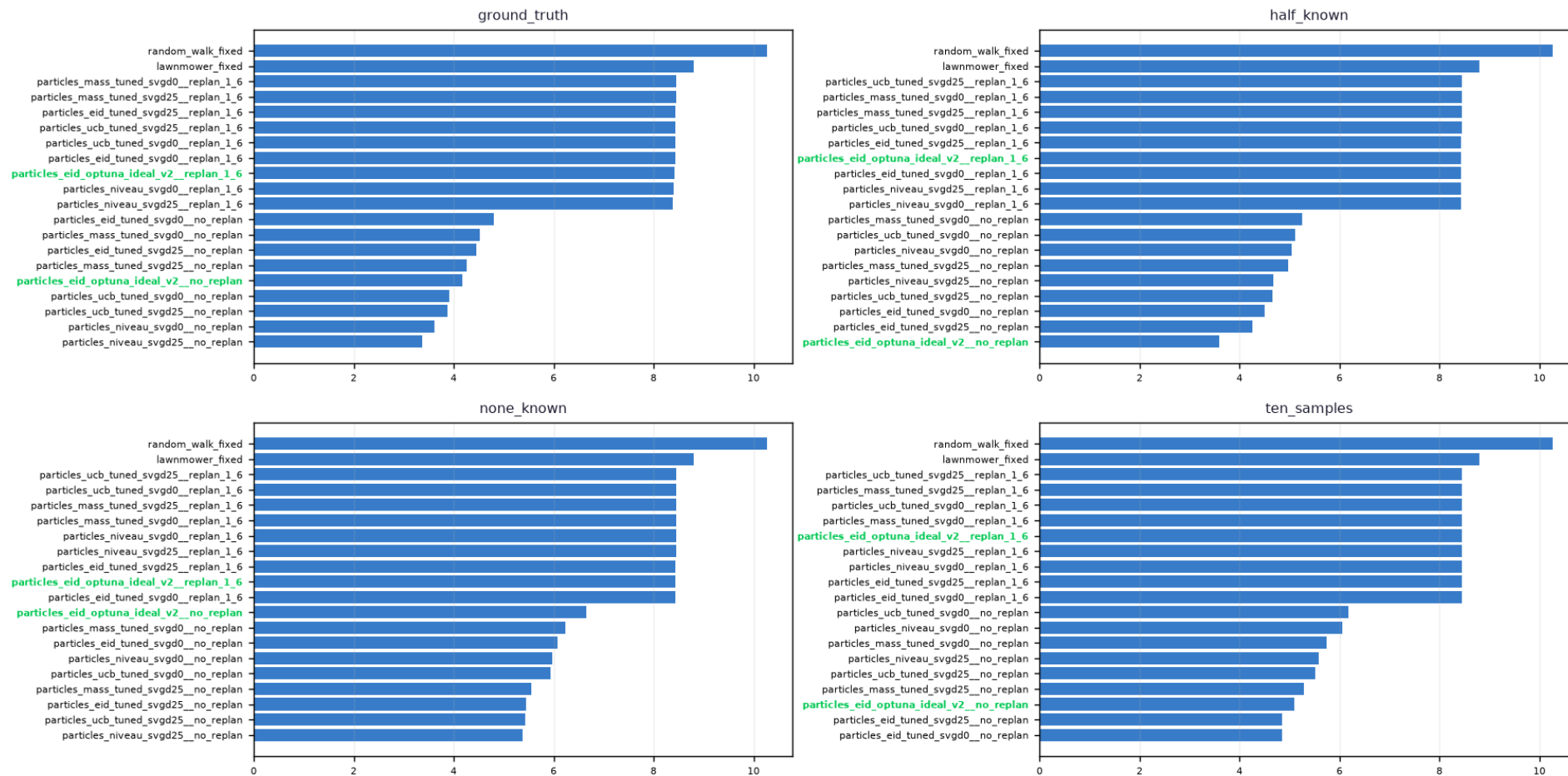


Evaluation pipeline

path_len

$$L(\gamma) = \sum_t \|\gamma(t+1) - \gamma(t)\|$$

no blanket better/worse -- a budget/cost quantity, not a quality metric



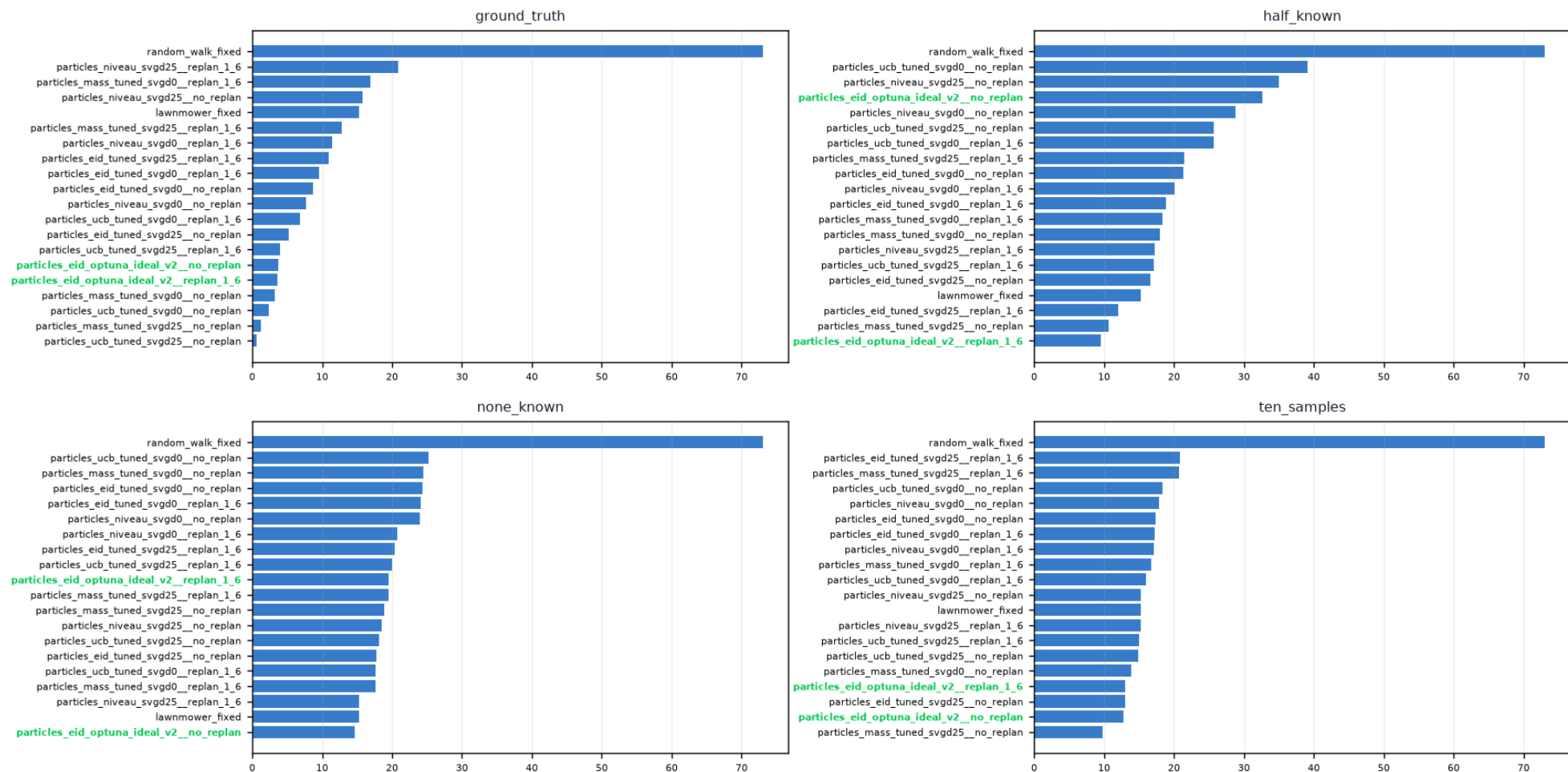
green bold label = Optuna ideal config (eid_optuna_ideal_v2)

Evaluation pipeline

$E_{\text{ergodic_total}}$

$$E_{\text{total}} = w \cdot \frac{1}{2} \sum_k \Lambda_k (c_k(\gamma) - \phi_k)^2$$

smaller = better



green bold label = Optuna ideal config (eid_optuna_ideal_v2)

Conclusion

- Optuna found results do seem to provide good results
- Very dependent on the setting
- Setting dependent configuration would be great
 - Further improve learning based configuration settings
 - For now: MLP with two hidden layers – supervised learning with candidates evaluated and labeled during the optimization
 - Next step: include J as unsupervised metric

3D Results

- Finished training on 200 different shapes (natural shapes e.g. different prisms, 3D letter versions, stanford library)
- Enhanced with se3 alignment constraint during inference

Thought looking forward

- Real Robot tests?
- Prepare a testing setup
- 3D robot arm trajectory constraints (constraints not just on surface) are so far not considered. Considering them would need a planning in joint space
- Improve MLP configuration prediction